# NGUYEN TAN HO

R&D Engineer in Deep Learning and Computer Vision

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1996/03/14

Passionate and self-motivated learner. My professional experience as an R&D engineer in academic environment with expertise in deep learning research and building prototypes to drive machine learning to solve real problems. As a recent Computer Science graduate, I am seeking an entry-level position to pursue a career as a Deep Learning Engineer/ MLOps Engineer.

## TECHNICAL SKILLS

**Programming languages | frameworks**
Python (SciKit-Learn, Pandas, NumPy), Matlab, Tensorflow, OpenCV

**Machine Learning:**
Classification, Clustering, Regression, CNN, Neural Network

**Web-API**
Flask, Django, HTML, CSS, JavaScript, Ajax, Streamlit

**Visualization**
Python (Matplotlib, Seaborn), Plotly.

## LANGUAGES

**English**
Working Proficiency

**Japanese**
Elementary

**Vietnamese**
Native

## EDUCATION

**MSc in Applied Informatic at HOSEI University** 2019/09 – 2021/09
GPA: 3.6/4.0 Tokyo, Japan

- A scientific paper got accepted and presented at a top conference (IEEE MMSP2021).
- Academic scholarships (Jasso, Daddy-Longleg) for students with good achievements.

**BSc in Computer Science at University of Information Technology** 2014/09 – 2019/06
Ho Chi Minh, Vietnam

- Thesis Grade: 9.8/10 - Top 1 highest grade in thesis defense UIT 2018
- Academic scholarship for Top 5% students of Academic Excellence 2018

## WORK EXPERIENCE

**R&D Engineer at Research Center for Computing & Multimedia Studies, HOSEI** 2021/03 – Present

PROJECT: MOEMO - Measuring Student Engagement in the Online Course

This project was developed to provide a solution of smart learning analytics system to measure student engagement in online environments.

**My role:**

- Led a small team and assigned appropriate tasks to team members.
- Implemented Deep learning-based methods such as Face recognition, Eye detection, Eye Gaze estimation, Emotion recognition to capture student features.
- Developed a prototype web application for the learning analysis using Flask RestfulAPI, SQLite3, Plotly.
- Co-oped with a data scientist to visualize the analytics results on the dashboard.

*(View demos in Portfolios)*

**Researcher at Intelligent Media Processing Laboratory, Hosei Univ.** 2020/12 – 2021/07

PROJECT: SpeedDeblur - A Framework to speed up CNN-based Deblurring for HEVC compressed video

This framework was designed and developed to reduce the processing time of Deep learning-based deblurring algorithms by up to 4.0× and 99.4× on GPU and CPU, respectively. This work was accepted by **IEEE MMSP 2021.**

**My role:**

- Proposed a novel strategy to generate blurry videos from the raw ones.
- Designed SpeedDeblur framework which can speed up the deblurring processing time by utilizing the coded information from the encoder.
- Implemented the state-of-the-art CNN-based deblurring algorithms from paper.
- Collected and processed data for training deblurring algorithms.
- Trained models and evaluated performance of SpeedDeblur compared with other state-of-the-art methods. (using PSNR, MSE metric).

*(View demos in Portfolios)*

**Research Assistant at Multimedia Communications Laboratory, VNUHCM-UIT** 2018/01 – 2018/07

PROJECT: A monitoring system to search people in surveillance camera network

This project was developed to provide a solution of the smart monitoring system to search people in surveillance camera networks. This work scored 9.8 for a bachelor thesis and got in the top 10 teams semifinal in Creative ideas for the smart city - Binh Duong Competition 2018, Vietnam.

**My role:**

- Implemented Machine Learning methods such as face detection, face recognition, face clustering, pedestrian detection to capture features.
- Trained VGG16 on the BETA dataset for person attribute recognition.
- Designed and developed a prototype web app for monitoring, searching and re-identifying people using face, cluster, or person information.
- Collected recorded video data and evaluated the system on face recognition.

*(View demos in Portfolios)*

## PUBLICATIONS

**SpeedDeblur: A Framework to speed up CNN-based Deblurring for HEVC compressed video**
Accepted by IEEE MMSP 2021
***Ho Tan Nguyen**, Chi Do-Kim Pham, Jinjia Zhou.*

**Briefing and Geovisualizing on International Practices of Learning Analytics in Higher Education**
Accepted by IEEE ICALT 2021
*H. Ueda, **Ho Tan Nguyen**, B. Huyen, T. Thuy, H. Hatakeyama and H. Nehal.*

**Can Sakai Log Data Improve Learning Analytics? Findings from a on Preliminary Survey**
Accepted by Information Processing Society of Japan
*H. Nehal, **Ho Tan Nguyen**, B. Huyen, T. Thuy, H. Hatakeyama, H. Ueda.*

**Students' emotion extraction and visualization for engagement detection in online learning**
Accepted by KES 2021
*H. Nehal, B. Huyen, T. Thuy, **Ho Tan Nguyen** ,Gökhan Akçapınar , H. Ueda.*

## CERTIFICATES

**Course Certificates of Neural Networks and Deep Learning**
by Coursera.org

**Course Certificates of Convolutional Neural Networks**
by Coursera.org